

Data Communications & Networking

Subject: Computer Networks (CS-401)

Unit: 01 (Introduction to OSI Model)

Date: June 13, 2026

1. Introduction to OSI Model

The **Open Systems Interconnection (OSI)** model is a conceptual framework developed by the International Organization for Standardization (ISO) in 1984. It describes how data from an application on one computer moves through a network medium to an application on another computer.

- It consists of **7 distinct layers**.
- Each layer performs a specific data communication task.
- Data flows from Layer 7 down to Layer 1 at the sender's side, and moves from Layer 1 up to Layer 7 at the receiver's side.

2. The Seven Layers (Top to Bottom)

Application Layer (Layer 7)

- **Purpose:** Directly interacts with the user or software application.
- **Functions:** Network virtual terminal, file transfer (FTAM), email services.
- **Protocols:** HTTP, HTTPS, FTP, SMTP, DNS.

Presentation Layer (Layer 6)

- **Purpose:** Manages the syntax and semantics of the information transmitted.
- **Functions:** Data translation, **Encryption/Decryption**, and Data Compression.

Session Layer (Layer 5)

- **Purpose:** Establishes, manages, and terminates sessions between applications.
- **Functions:** Dialog control, token management, and synchronization (checkpoints).

Transport Layer (Layer 4)

- **Purpose:** Ensures reliable end-to-end data delivery.
- **Functions:** Service-point addressing (Port numbers), Segmentation and reassembly, Connection control, Flow control, and Error control.
- **Protocols:** TCP (Transmission Control Protocol), UDP (User Datagram Protocol).

Network Layer (Layer 3)

- **Purpose:** Source-to-destination delivery of packets across multiple networks.
- **Functions:** **Logical Addressing (IP Routing)** and Routing (finding the best path).
- **Devices:** Routers, Layer 3 Switches.

Data Link Layer (Layer 2)

- **Purpose:** Node-to-node delivery of data.
- **Functions:** Framing, Physical addressing (MAC address), Flow control, and Error control (CRC).
- **Devices:** Switches, Bridges.

Physical Layer (Layer 1)

- **Purpose:** Transmitting raw bits over a physical medium.
- **Functions:** Bit synchronization, Bit rate control, Line configuration, Physical topology.
- **Medium:** Coaxial cable, Fiber optics, Twisted-pair wire.

3. Summary Table: Data Units and Devices

Layer Name	Data Unit (PDU)	Primary Devices/Protocols
Application	Data	HTTP, FTP, SMTP
Presentation	Data	SSL, TLS, ASCII
Session	Data	NetBIOS, RPC
Transport	Segments (TCP) / Datagrams (UDP)	TCP, UDP
Network	Packets	Routers, IPv4/IPv6
Data Link	Frames	Switches, MAC Address
Physical	Bits	Hubs, Cables, Repeaters

Quick Revision Mnemonic (Layer 1 to 7):

*"Please Do N**ot **Throw **S**ausage **P**izza **A**way"*

(Physical, Data Link, Network, Transport, Session, Presentation, Application)